

Advanced (Habanero)

- Q1.** Which of the following is equivalent to the ratio 3 : 5?
- (A) 6 : 10
(B) 9 : 15
(C) 1 : 2
(D) 2 : 3
(E) 4 : 5
- Q2.** Solve the proportion $\frac{2}{3} = \frac{x}{6}$ using cross multiplication.
- (A) $x = 2$
(B) $x = 3$
(C) $x = 4$
(D) $x = 5$
(E) $x = 6$
- Q3.** A map has a scale of 1 : 1000. If a distance on the map is 5 cm, what is the actual distance?
- (A) 5 m
(B) 50 m
(C) 500 m
(D) 5000 m
(E) 50000 m
- Q4.** A recipe calls for a ratio of 2 : 3 of flour to sugar. If you use 4 cups of flour, how many cups of sugar should you use?
- (A) 4 cups
(B) 5 cups
(C) 6 cups
(D) 8 cups
(E) 10 cups
- Q5.** A bicycle travels 25 miles in 2 hours. What is the ratio of distance to time?
- (A) 10 : 1
(B) 12.5 : 1
(C) 15 : 1
(D) 20 : 1
(E) 25 : 1
- Q6.** A group of friends want to share some candy in a ratio of 3 : 2. If they have 60 pieces of candy, how many pieces will each person get?
- (A) 10 pieces
(B) 12 pieces
(C) 15 pieces
(D) 18 pieces
(E) 20 pieces
- Q7.** A water tank can hold 1200 liters of water. If 300 liters of water are already in the tank, what is the ratio of water in the tank to the total capacity?
- (A) 1 : 2
(B) 1 : 3
(C) 1 : 4
(D) 1 : 5
(E) 1 : 6
- Q8.** A car travels 250 miles in 5 hours. What is the ratio of distance to time?
- (A) 40 : 1
(B) 50 : 1
(C) 60 : 1
(D) 70 : 1
(E) 80 : 1

Open Ended Questions (FRQ)

- Q9.** A bakery sells a total of 480 loaves of bread per day. They sell a combination of whole wheat and white bread in a ratio of 3 : 5. How many loaves of each type of bread are sold per day? Show your work and explain your answer.
- Q10.** A farmer has 150 acres of land to plant crops. He wants to plant a combination of corn and soybeans in a ratio of 2 : 3. If he plants 40 acres of corn, how many acres of soybeans will he plant? Show your work and explain your answer.

Chef's Tips

- Emphasize the importance of labeling and units in ratio and proportion problems.
- Encourage students to use cross multiplication to solve proportions.
- Remind students to check their work and explain their answers clearly.